

Entropy of Discrete Information Source

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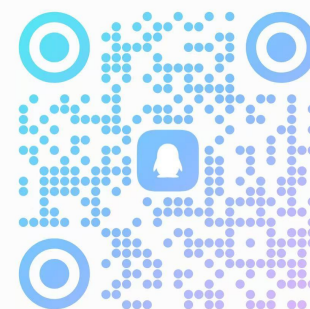
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信息论基础-20...

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Why Quantify Discrete Info. Source?



Which game generates more valuable information?

We need to quantify the game itself!

Main Content

1 Entropy of Discrete Info. Source

2 Conditional and Joint Entropy

3 Properties of Entropy

Entropy of Discrete Info. Source

$$\begin{pmatrix} X \\ P(X) \\ I(X) \end{pmatrix} = \begin{pmatrix} a_1 & \dots & a_i & \dots & a_n \\ p(a_1) & \dots & p(a_i) & \dots & p(a_n) \\ I(a_1) & \dots & I(a_i) & \dots & I(a_n) \end{pmatrix}$$

Entropy is defined as the average self-information of all possible messages sent by the discrete information source.

$$H(X) = \sum_{i=1}^n p(a_i) \cdot I(a_i) = - \sum_{i=1}^n p(a_i) \log_2 p(a_i)$$

Examples – Football Game

Discrete Info.

Source X :

2018-2019 ECL Semi 1st Round:

Barcelona (0) vs Liverpool (0)

Barcelona goes to the final with 58.5%

Liverpool goes to the final with 41.5%

$$H(X) = -58.5\% \log_2 58.5\% - 41.5\% \log_2 41.5\% = 0.9791 \text{ bit/sign}$$

Discrete Info.

Source Y :

2018-2019 ECL Semi 2nd Round:

Liverpool (0) vs Barcelona (3)

Barcelona goes to the final with 99.9%

Liverpool goes to the final with 0.01%

$$H(Y) = -99.9\% \log_2 99.9\% - 0.1\% \log_2 0.1\% = 0.0114 \text{ bit/sign}$$

Example -- Weather

$$\begin{pmatrix} \text{Weather} \\ \text{Prob. Space} \end{pmatrix} = \begin{pmatrix} \text{Sunny} & \text{Cloudy} & \text{Rainy} & \text{Snowy} \\ 1/2 & 1/4 & 1/8 & 1/8 \end{pmatrix}$$

Solution:

$$\begin{aligned} H(X) &= -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{4} \log_2 \frac{1}{4} - \frac{1}{8} \log_2 \frac{1}{8} - \frac{1}{8} \log_2 \frac{1}{8} \\ &= 1.75 \text{ bit} \end{aligned}$$

Example

Assume a discrete memoryless information source with distribution

X	$x_1 = 0$	$x_2 = 1$	$x_3 = 2$	$x_4 = 3$
$P(X)$	$3/8$	$1/4$	$1/4$	$1/8$

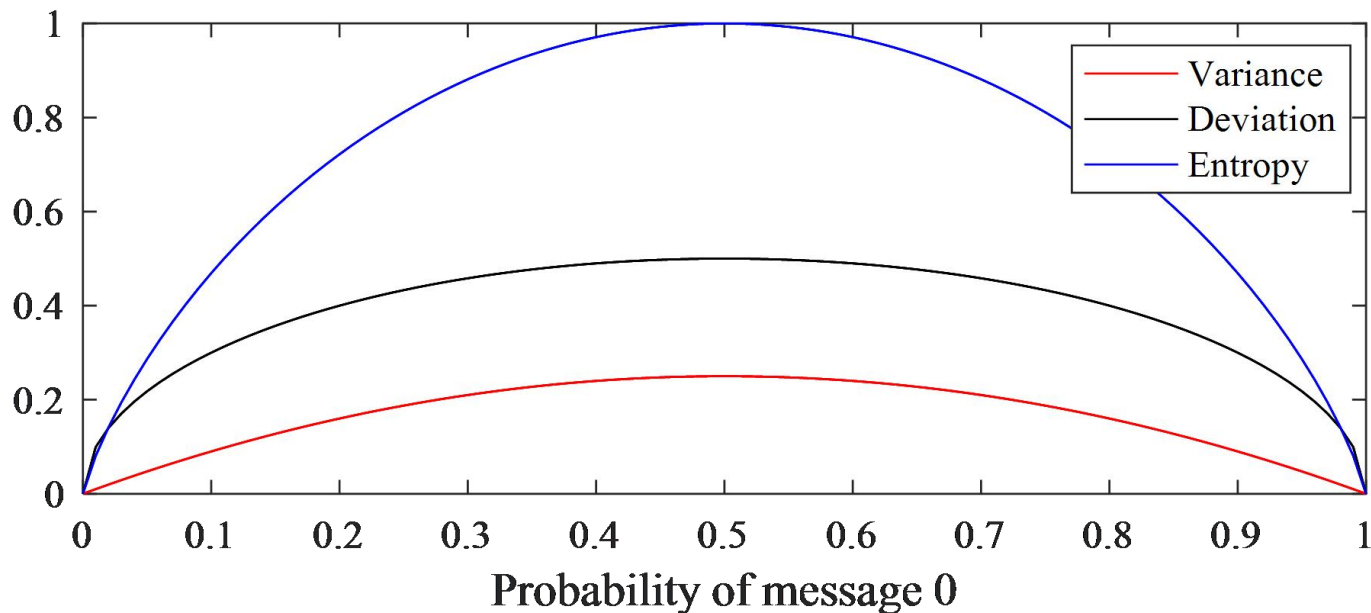
The message emitted by this source is:

(202120130213001203210110321010021032011223210),

Find:

1. What is the **average amount of information per symbol** carried by **this message**?
2. What is the **average amount of information per symbol** carried by **this source**?

Three physical meanings of entropy



□ The entropy $H(X)$ represents:

- The average information carried by each message.
- The average uncertainty of the information source
- The randomness of the random variable, similar to variance and standard deviation.

Joint Entropy

□ Mathematical model of the joint discrete information source XY :

$$\begin{pmatrix} XY \\ P(XY) \end{pmatrix} = \begin{pmatrix} a_1b_1, \dots a_1b_m, a_2b_1, \dots a_2b_m, \dots a_nb_1, \dots a_nb_m \\ p(a_1b_1), \dots p(a_1b_m), p(a_2b_1), \dots p(a_2b_m), \dots p(a_nb_1), \dots p(a_nb_m) \end{pmatrix}$$

□ Joint entropy is defined as the expectation of the information carried by every joint message a_ib_j sent by the joint information source XY :

$$\begin{aligned} H(XY) &= E[I(a_ib_j)] = \sum_{j=1}^m \sum_{i=1}^n p(a_ib_j) I(a_ib_j) \\ &= - \sum_{j=1}^m \sum_{i=1}^n p(a_ib_j) \log_2 p(a_ib_j) \end{aligned}$$

Conditional Entropy – Loss Entropy

□ Mathematical model of the joint discrete information source XY :

$$\begin{pmatrix} XY \\ P(XY) \end{pmatrix} = \begin{pmatrix} a_1b_1, \dots a_1b_m, a_2b_1, \dots a_2b_m, \dots a_nb_1, \dots a_nb_m \\ p(a_1b_1), \dots p(a_1b_m), p(a_2b_1), \dots p(a_2b_m), \dots p(a_nb_1), \dots p(a_nb_m) \end{pmatrix}$$

□ Given the information destination Y , the conditional entropy $H(X|Y)$ of the information source X can be expressed as:

$$H(X|Y) = E[I(a_i|b_j)] = \sum_{j=1}^m \sum_{i=1}^n p(a_ib_j) I(a_i|b_j)$$

**Channel Equivocation,
Loss Entropy**

$$= - \sum_{j=1}^m \sum_{i=1}^n p(a_ib_j) \log_2 p(a_i|b_j)$$

Conditional Entropy – Noise Entropy

□ Mathematical model of the joint discrete information source XY :

$$\begin{pmatrix} XY \\ P(XY) \end{pmatrix} = \begin{pmatrix} a_1b_1, \dots a_1b_m, a_2b_1, \dots a_2b_m, \dots a_nb_1, \dots a_nb_m \\ p(a_1b_1), \dots p(a_1b_m), p(a_2b_1), \dots p(a_2b_m), \dots p(a_nb_1), \dots p(a_nb_m) \end{pmatrix}$$

□ Given the information source X , the conditional entropy $H(Y/X)$ of the information destination Y can be expressed as:

$$H(Y|X) = E[I(b_j|a_i)] = \sum_{j=1}^m \sum_{i=1}^n p(a_i b_j) I(b_j|a_i)$$



Noise Entropy

$$= - \sum_{j=1}^m \sum_{i=1}^n p(a_i b_j) \log_2 p(b_j|a_i)$$

Example

- Suppose there are 100 balls in a box, including 40 black balls and 60 white balls. You draw balls without replacement, and perform the experiment twice in a row. Find the difficulty (i.e., the probability/information measure, depending on the context) of correctly guessing the color of the second ball drawn

Non-negative, Symmetric, Certainty

- ❑ **Non-negative:** The entropy of the information source is the expectation of the information carried by all the possible messages. Since the self-information is non-negative, so is the entropy.
- ❑ **Symmetric:** Changing the order of the messages' probabilities during the calculation may not alter the value of the entropy.

$$H(X) = H[p(a_1), \dots, p(a_n)] = - \sum_{i=1}^n p(a_i) \log_2 p(a_i)$$

The entropy only depends on the overall statistics of the information source. It is not relied upon on some specific messages, or the specific value of the messages.

- ❑ **Certainty:** If the probability of the source generating a message is 1, its entropy is zero. This is a certain source and its uncertainty is 0

$$H(1,0) = H(1,0, \dots, 0) = H(0,1, \dots, 0) = \dots = 0$$

A detailed illustration of the character Wukong from the game Black Myth: Wukong. He is depicted as a monkey with a fierce expression, wearing ornate, dark blue and gold armor. He is holding a sword in his right hand. The background is dark and features faint, traditional Chinese ink wash paintings of figures.

BLACK MYTH WUKONG

<https://gl.ali213.net/html/2024-8/1484815.html>

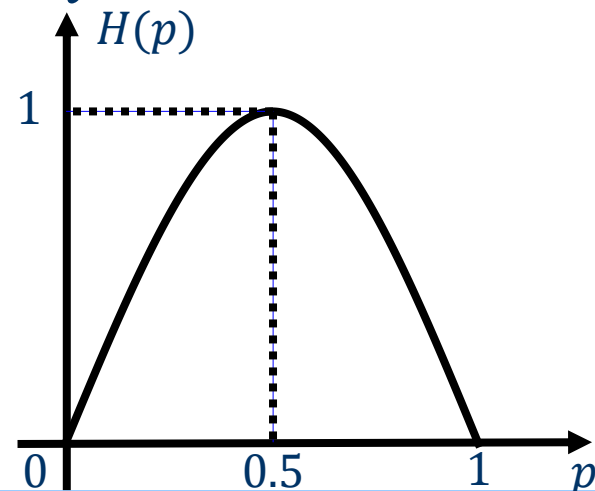
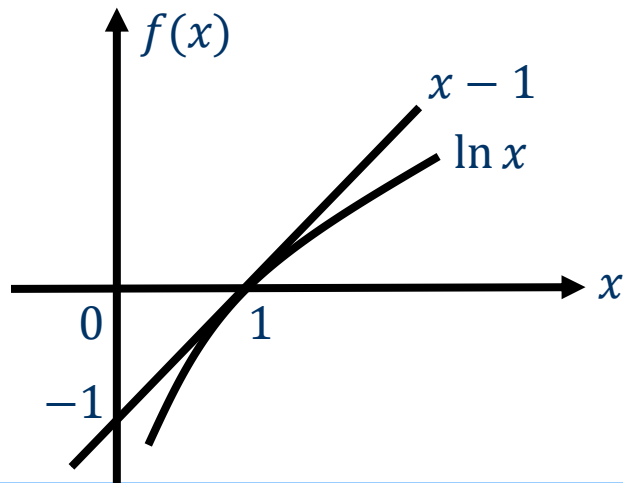
Maximum Entropy of Discrete Source

□ **Theorem:** A information source X may generate n possible messages, its entropy should satisfy the following inequality:

$$H(X) \leq \log_2 n$$

When all the n messages are generated by the source X with equal probabilities, the equality holds.

Proof: By exploiting the property of the natural logarithm $\ln x \leq x - 1$, $x > 0$, when we have $x = 1$, the equality holds.



The Entropy is Extendable

- A information source X may generate n possible messages, while another source Y may generate $(n + 1)$ messages. If a message is generated by Y with a probability tending to zero but the other messages have almost the same probabilities with those generated by X , the entropy of Y and that of X are the same, which are expressed as

$$\begin{aligned} & \lim_{\varepsilon \rightarrow 0} H_{n+1}[p(a_1), \dots, p(a_n) - \varepsilon, p(a_{n+1}) = \varepsilon] \\ &= H_n[p(a_1), \dots, p(a_n)] \end{aligned}$$

This equation holds, since we have $\lim_{\varepsilon \rightarrow 0} \varepsilon \log_2 \varepsilon = 0$.

- Although a message having a small probability carry high information, but it contribute a little to the entropy of the information source.
- In the case of having s additional messages, whose probabilities are tending to zeros, the similar expansible properties also hold.

The Entropy is Additive

- Given two random information source X and Y , their joint entropy, their individual entropies and their conditional entropies satisfy the following equations:

$$H(XY) = H(X) + H(Y/X)$$

$$H(XY) = H(Y) + H(X/Y)$$

This is the additive properties of the entropy.

Proof: Exploit the definitions of the entropy, conditional entropy and joint entropy, as well as the probabilistic properties of the random variables to prove the additive property.

- When information sources X and Y are **independent** of each other, their joint entropy can be the summation of their individual entropies:

$$H(XY) = H(X) + H(Y)$$

Extremum of the Entropy

- For an arbitrary pair of random information source X and Y , the following inequality holds:

$$\begin{aligned} H_n[p(a_1), \dots, p(a_n)] &= - \sum_{i=1}^n p(a_i) \log_2 p(a_i) \\ &\leq - \sum_{i=1}^n p(a_i) \log_2 p(b_i) \end{aligned}$$

For an arbitrary probability distribution $\{p(a_i)\}$, taking the average of another probability distribution $\{p(b_i)\}$'s self-information on it, the resultant value is higher than its own entropy.

- By exploiting the extremum of the entropy, we can prove that the **conditional entropy is lower than the unconditional one**:

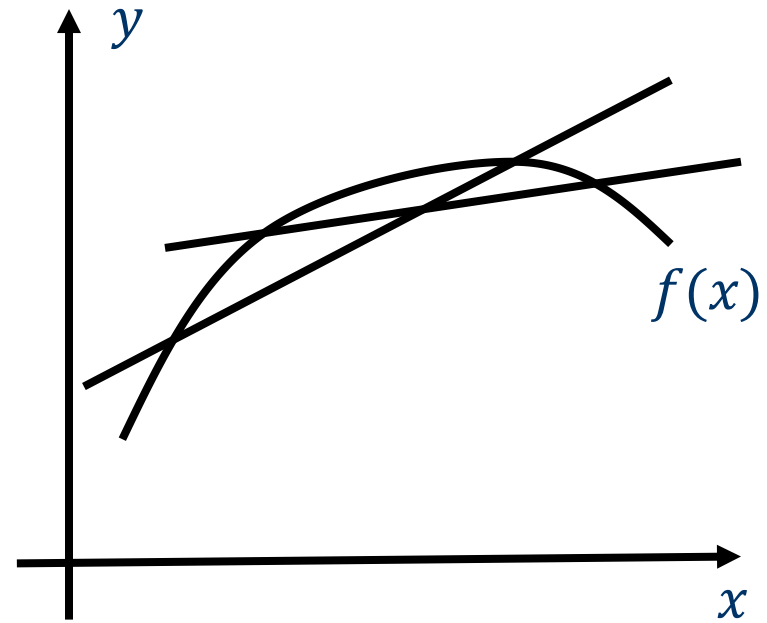
$$H(X/Y) \leq H(X)$$

Concavity of the Entropy

- For a multi-variable function $f(a_1, \dots, a_n) = f(\mathbf{X})$, given an arbitrary positive number α and an arbitrary pair of vectors $\mathbf{X}$ and $\mathbf{Y}$ in its definitional domain, if we have

$$f[\alpha\mathbf{X} + (1 - \alpha)\mathbf{Y}] \geq \alpha f(\mathbf{X}) + (1 - \alpha)f(\mathbf{Y}),$$

$f(\mathbf{X})$ is then a concave function.



- The entropy is a **concave** function. Given a pair of probability vectors $\mathbf{P} = [p(a_1), \dots, p(a_n)]$ and $\mathbf{Q} = [p(b_1), \dots, p(b_n)]$, where we have $0 \leq p(a_i) \leq 1$, $0 \leq p(b_i) \leq 1$ and $\sum_{i=1}^n p(a_i) = \sum_{i=1}^n p(b_i) = 1$, the following inequality holds for an arbitrary positive number α :

$$H[\alpha\mathbf{P} + (1 - \alpha)\mathbf{Q}] \geq \alpha H(\mathbf{P}) + (1 - \alpha)H(\mathbf{Q})$$

Example

□ Given an arbitrary random variable X and its function $g(X)$, prove that the entropy $H[g(X)]$ of the function $g(X)$ is not higher than the entropy $H(X)$ of X by obeying the following steps.

$$H[X, g(X)] \stackrel{(a)}{=} H(X) + H[g(X)/X]$$

$$\stackrel{(b)}{=} H(X)$$

$$H[X, g(X)] \stackrel{(c)}{=} H(g(X)) + H[X/g(X)]$$

$$\stackrel{(d)}{\geq} H[g(X)]$$

Finally, we may obtain the inequality $H[g(X)] \leq H(X)$.

Example

- Assume that the joint probabilities of the random variables X and Y are given by the following table:

$X \backslash Y$	0	1
0	$\frac{1}{3}$	$\frac{1}{3}$
1	0	$\frac{1}{3}$

Calculate $H(X)$, $H(Y)$, $H(X/Y)$, $H(Y/X)$, $H(X, Y)$, $H(X) - H(X/Y)$ and $H(Y) - H(Y/X)$.

Why Quantify Discrete Info. Source?

Discrete Info. **2018 World Cup Group H: Germany vs Korea**
Source X : **Germany wins 81.3%, Draw 13%, Korea wins 5.7%**

Discrete Info. **2018 World Cup Final (90 mins): France vs Croatia**
Source Y : **France wins 50%, Draw 29%, Croatia wins 21%**

**Which game generates more valuable
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We need to quantify the game itself!

Examples – Football Game

Discrete Info. 2018 World Cup Group H: Germany vs Korea

Source X : Germany wins 81.3%, Draw 13%, Korea wins 5.7%

$$H(X) = -81.3\% \log_2 81.3\% - 13\% \log_2 13\%$$

$$-5.7\% \log_2 5.7\% = 0.86 \text{ bit}$$

Discrete Info. 2018 World Cup Final (90 mins): France vs Croatia

Source Y : France wins 50%, Draw 29%, Croatia wins 21%

$$H(Y) = -50\% \log_2 50\% - 29\% \log_2 29\%$$

$$-21\% \log_2 21\% = 1.5 \text{ bit}$$